Turkish NLP Benchmarking LLMs

Course: CSE4078S26 – Group 9

Gemma 2B IT vs Qwen3 1.7B

Dataset: alibayram/doktorsitesi

Team & Repository Setup

Git Architecture

- **src/:** Core inference scripts for model evaluation
- **tests/:** Unit tests and evaluation benchmarks
- **docs/:** Project documentation and asset reporting

Workflow Status

Modular scripts for inference and evaluation are fully operational. All results are currently exported as structured JSON summaries for automated reporting and reproducibility.

Dataset Inspection

Source Profile

- **Library:** Hugging Face (Streaming Mode)
- **Content:** Professional Turkish Medical Q&A
- **Primary Fields:**
  - *question_content*
  - *question_answer*
  - *doctor_specialty*



Structured Medical Data

First Evaluation Results

2,000

SAMPLES EVALUATED

Gemma 2B IT: 12.4 PPL

Qwen3 1.7B: 18.2 PPL

Gemma currently leads by **5.8 points** in linguistic coherence.

Defining Perplexity (PPL)

The Metric

Perplexity measures how well a probability model predicts a sample. It is the exponentiated average negative log-likelihood of a sequence.

```
PPL(W) = P(w1, w2, ..., wN)^(-1/N)
```

The Intuition

Think of it as **"How surprised is the model by the test data?"**

- **Low PPL:** The model expected these words; high confidence in language patterns.
- **High PPL:** The model was "surprised" and struggled to predict expert answers.

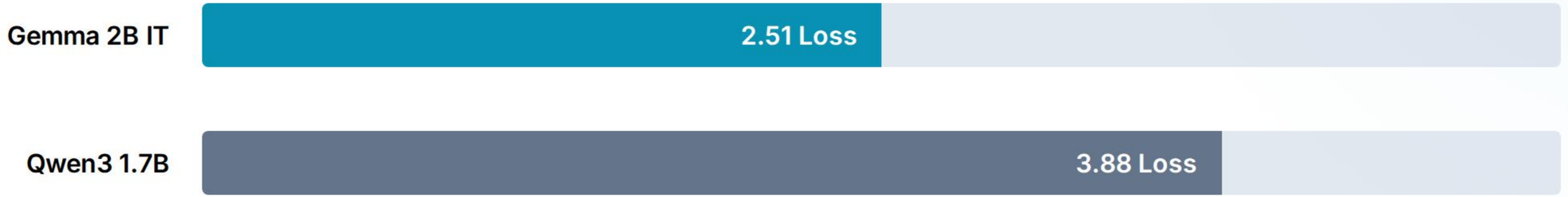
Interpretation: What results show

Clinical Linguistic Alignment

The performance gap suggests several critical findings for our pipeline:

- **Domain Familiarity:** Gemma's pre-training captured Turkish medical terminology more effectively.
- **Next-Token Fit:** Higher probability of generating Ground Truth expert medical answers.
- **Foundation Strength:** Gemma is the superior candidate for the upcoming LoRA fine-tuning phase.

Loss Comparison Analysis



**Computed across 2,000 context-masked samples. Lower loss indicates better next-token alignment with professional interaction styles.*

Preprocessing & Engineering

- **Chat Templates:** Implemented model-specific formatting to ensure consistent prompting across architectures.
- **4-bit Quantization:** Loaded models via BitsAndBytes for efficient testing on standard hardware.
- **Token Masking:** Context tokens are ignored during loss computation to focus strictly on answer accuracy.

Predictive Task Formulation

Task A: Specialty

Goal: Predict medical specialty from patient query text.

Metrics: Accuracy, F1-Score

Task B: Quality

Goal: Predict answer perplexity bands as a proxy for answer style matching.

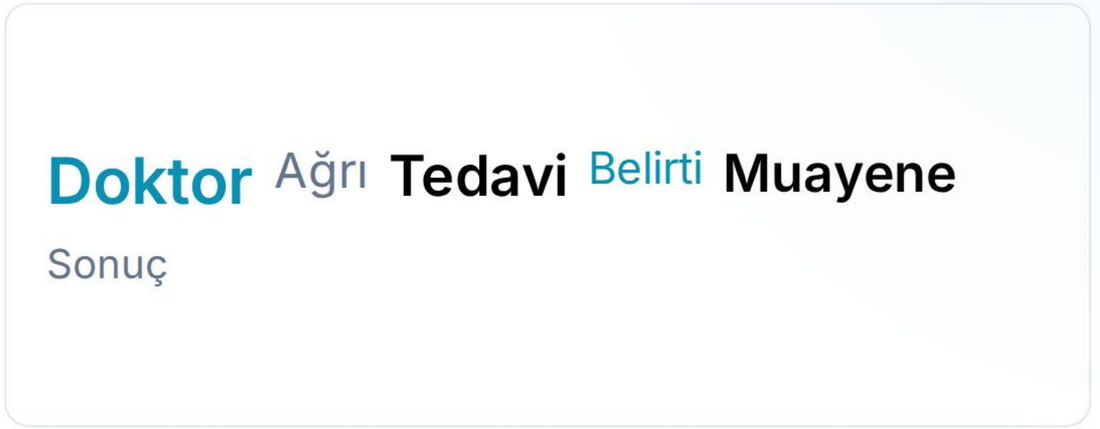
Metrics: RMSE, Calibration

Exploratory Data Analysis

Linguistic Observations

Detected high variation in patient question lengths compared to standardized expert responses.

Class density peaks in Dermatology (22%) and Internal Medicine (18%).



Remaining Work Plan

1

Scaling

Broader benchmarks & samples

2

LoRA

Targeted fine-tuning

3

Analysis

Error breakdown by specialty

4

Reporting

Reproducible final results

Key Takeaways

The project pipeline is operational end-to-end. Current benchmarks identify Gemma as the superior baseline, providing a solid foundation for the upcoming fine-tuning and error analysis phase.